

WILLIAM O'CONNELL

Mechatronics Engineering Student

PROFILE

Mechatronics Engineering student at the University of Auckland with a track record of turning ideas into working hardware - from a self-built wind tunnel to an automated irrigation system. Drawn to designing products that make a real difference to people's lives, and keen to bring hands-on prototyping skills, cross-disciplinary teamwork, and a fast-learning curve to a product development or process engineering internship.

PROJECTS

Wind Tunnel | Personal Project

2026

- Designed and manufactured a functional wind tunnel from scratch to study airflow effects on objects such as car bodies.
- Constructed out of plywood and acrylic, using a 12V PC fan for airflow. Reaches a top speed of approximately 10 metres per second.
- Used Autodesk Fusion to CAD the design, as well as Arduino IDE for coding.
- Full project write-up and additional builds available in portfolio: [William O'Connell — Mechatronics Engineering](#)

ESD Community Project — Automated Irrigation System | UOA Engineers for Sustainable Development

2026

- Contributed to the design and build of an automated irrigation system as part of a student engineering-for-good initiative. Worked on fluid calculations and part selection.

EDUCATION

Bachelor of Engineering (Honours), Mechatronics Engineering | University of Auckland

March 2025 – Present

- Cumulative GPA: 7.83
- Coursework: Engineering Mechanics, Mathematical Modelling, Electrical & Digital Systems, Principles of Engineering Design, Materials Science, Introduction to Mechatronics, Design & Manufacture

NCEA Level 3 | Sacred Heart College, Auckland

January 2018 – November 2024

- Achieved NCEA Level 3 with Excellence

WORK EXPERIENCE

Tutor | MyTuition, Auckland

March 2026 – Present

- Tutor students both online and in person, breaking down technical concepts clearly.

Front of House | Burger Burger, Auckland

January 2024 – May 2025

- Delivered consistent customer service in a fast-paced team environment, handling orders, payments, and problem-solving under time pressure.

Front of House | NZ Bakery, Auckland

January 2022 – March 2023

- Handled transactions and inventory accurately while maintaining a high standard of service.

CONTACT

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[William O'Connell — Mechatronics Engineering](#)

SKILLS

Aerodynamics & CFD exposure

Autodesk Fusion and SolidWorks

MATLAB and C

Embedded Systems

Cross-disciplinary collaboration

AWARDS

- Solvett Finalist (2025) — entrepreneurial group project; won the Eden Park Project using IoT for wayfinding
- Proxime Accessit, Sacred Heart College (2024)
- University of Auckland 140th Anniversary Scholarship
- NCEA Scholarship Award (2024)

CO-CURRICULAR

UOA Rocketry Club

UOA Engineers for Social Development

UOA Spanish Society & Latino Society

UOA Engineering Running Club

INTERESTS

Running & gym - completed several half marathons and the 2026 Hawke's Bay Marathon

Motorsport, sailing, football, cricket

Camping, hiking, and the outdoors

Guitar

LANGUAGES

English

Spanish (B2)